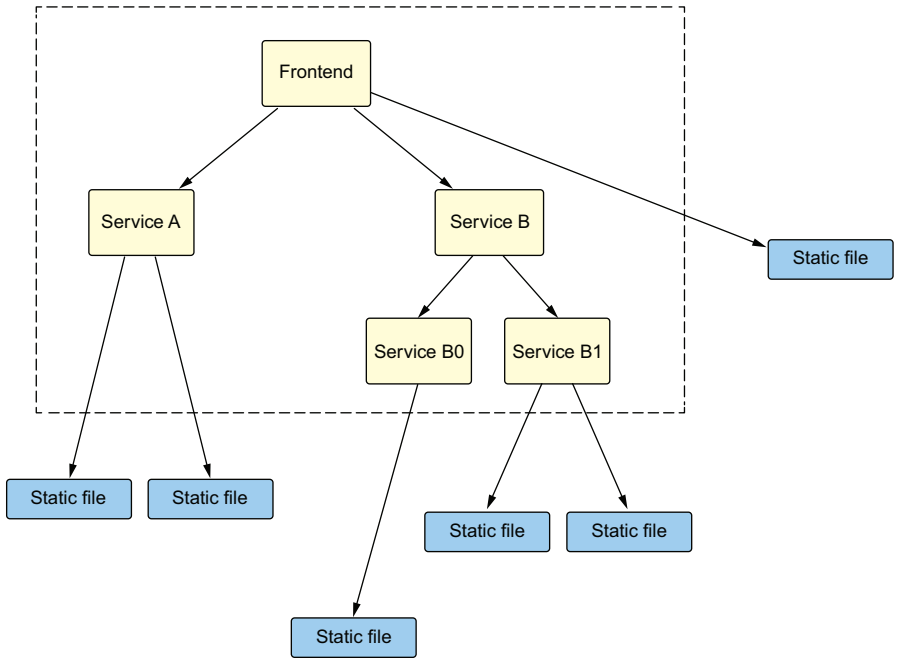


Figure 8.4 Example microservices app architecture

services into the distributed tracing tool (CNCF Jaeger). The setup is based on what you already saw in chapter 6.

8.2.2 Implementing the example app

To stand up our example app stack, using the Docker Compose file shown in listing 8.1, execute the following command from within the `ch08/jaeger/` directory:

```
docker-compose -f docker-compose.yaml up
```

> **NOTE** The ho11y synthetic signal generator has several configuration options that are defined via environment variables. With `HO11Y_PORT`, you can define on which port `ho11y` should be listening for service requests, and with `DOWNSTREAM0` to `DOWNSTREAM4`, you can define URLs that the `ho11y` instance should call (and, hence, allow you to wire up and simulate arbitrary complex microservices). To simulate dropping requests, we will use the `HO11Y_INJECT_FAILURE` environment variable, which causes the `ho11y` instance to return some `4xx` or `5xx` HTTP status codes in half of the cases, rather than returning a 200 HTTP status code. Finally, to simulate another kind failure mode (throttling of the upstream service), we will use the `HO11Y_CUTOFF_TPS` environment variable to define a cutoff point, beyond which `ho11y` will return a 429 HTTP status code (https://httpstatuscodes.org/429/).